$$> f(b,c,r) := \left(\frac{1}{2} - \cos\left(\frac{5}{6}\pi - c - r\right)\cdot\sin(c) - \cos\left(\frac{5}{6}\pi - b + r\right)\cdot\sin(b)\right)^2 + \left(\sin\left(\frac{5}{6}\pi - c - r\right)\cdot\sin(c) - \sin\left(\frac{5}{6}\pi - b + r\right)\cdot\sin(b)\right)^2$$

$$f := (b,c,r) \mapsto \left(\frac{1}{2} - \cos\left(\frac{5\cdot\pi}{6} - c - r\right)\cdot\sin(c) - \cos\left(\frac{5\cdot\pi}{6} - b + r\right)\cdot\sin(b)\right)^2 + \left(\sin\left(\frac{5\cdot\pi}{6} - c - r\right)\cdot\sin(c) - \sin\left(\frac{5\cdot\pi}{6} - b + r\right)\cdot\sin(b)\right)^2 \quad \textbf{(1)}$$

$$> g(b,c,r) := simplify\left(expand\left(f(b,c,0)^2 - f(b,c,r)\cdot f(b,c,-r)\right)\right)$$

$$g := (b,c,r) \mapsto simplify\left(expand\left(f(b,c,0)^2 - f(b,c,r)\cdot f(b,c,-r)\right)\right) \quad \textbf{(2)}$$

$$> collect(expand(g(b,c,r)), \cos(r))$$

$$\left(\sin(c)\sqrt{3}\cos(c)\sin(b)^2 + \sin(c)^2\sin(b)\sqrt{3}\cos(b) + \cos(c)^2\cos(b)^2 - \sin(c)\cos(c)\sin(b)\cos(b) - 1\right)\cos(r)^2 + \left(-3\cos(b)^4 + \frac{7\cos(b)^2}{4} + 4\cos(b)^4\cos(c)^2 + 4\cos(b)^2\cos(c)^4 - 8\cos(c)^2\cos(b)^2 + \sqrt{3}\sin(b)\cos(b)^3 - \frac{13\sin(b)\sqrt{3}\cos(b)}{4} + \sin(c)\sqrt{3}\cos(c)^3 - \frac{13\sin(c)\sqrt{3}\cos(c)}{4} + 2\sqrt{3}\sin(b)\cos(b)\cos(c)^2 + 8\sin(c)\cos(c)\sin(b)\cos(b) - 4\sin(c)\cos(c)\sin(b)\cos(b)^3 + 2\sin(c)\sqrt{3}\cos(c)\cos(b)^2 - 4\sin(c)\cos(c)^3\sin(b)\cos(b) + \frac{5}{2} - 3\cos(c)^4 + \frac{7\cos(c)^2}{4}\right)\cos(r) + \frac{9\sin(c)\sqrt{3}\cos(c)}{4} - \sqrt{3}\sin(b)\cos(b)^3 + \frac{9\sin(b)\sqrt{3}\cos(b)}{4} - \sin(c)\sqrt{3}\cos(c)^3 + 4\sin(c)\cos(c)\sin(b)\cos(b)^3 - \sin(c)\sqrt{3}\cos(c)\cos(b)^2 + 4\sin(c)\cos(c)^3\sin(b)\cos(b) - \sqrt{3}\sin(b)\cos(b)\cos(c)^2 - 7\sin(c)\cos(c)\sin(b)\cos(b) - \frac{7\cos(c)^2}{4} + 3\cos(c)^4 - 4\cos(b)^2\cos(c)^4 + 7\cos(c)^2\cos(b)^2 + 3\cos(b)^4 - \frac{7\cos(b)^2}{4} - 4\cos(b)^4\cos(c)^2 - \frac{3}{2} \quad \textbf{(3)}$$

$$> simplify\Bigg(\frac{1}{r^2}\Bigg(\left(\sin(c)^2\sin(b)\sqrt{3}\cos(b) + \sin(c)\sqrt{3}\cos(c)\sin(b)^2 - \sin(c)\cos(c)\sin(b)\cos(b) + \cos(b)^2\cos(c)^2 - 1\right)\cdot\left(1 - r^2\right) + \Bigg($$

$$-4\sin(c)\cos(c)\sin(b)\cos(b)^3+2\sin(c)\sqrt{3}\cos(c)\cos(b)^2$$

$$+2\sqrt{3}\sin(b)\cos(b)\cos(c)^2+8\sin(c)\cos(c)\sin(b)\cos(b)$$

$$-4\sin(c)\cos(c)^3\sin(b)\cos(b)-3\cos(c)^4+\frac{7\cos(c)^2}{4}-3\cos(b)^4+\frac{7\cos(b)^2}{4}$$

$$+\frac{5}{2}+\sin(c)\sqrt{3}\cos(c)^3-\frac{13\sin(c)\sqrt{3}\cos(c)}{4}+\sqrt{3}\sin(b)\cos(b)^3$$

$$-\frac{13\sin(b)\sqrt{3}\cos(b)}{4}-8\cos(b)^2\cos(c)^2+4\cos(b)^2\cos(c)^4+4\cos(c)^2\cos(b)^4\Bigg)$$

$$\cdot\left(1-\frac{r^2}{2}\right)+4\sin(c)\cos(c)\sin(b)\cos(b)^3-7\sin(c)\cos(c)\sin(b)\cos(b)$$

$$-\sin(c)\sqrt{3}\cos(c)\cos(b)^2-\sqrt{3}\sin(b)\cos(b)\cos(c)^2+4\sin(c)\cos(c)^3\sin(b)\cos(b)$$

$$-\frac{7\cos(b)^2}{4}-\frac{7\cos(c)^2}{4}+3\cos(c)^4+3\cos(b)^4-\sin(c)\sqrt{3}\cos(c)^3$$

$$+\frac{9\sin(c)\sqrt{3}\cos(c)}{4}-\sqrt{3}\sin(b)\cos(b)^3+\frac{9\sin(b)\sqrt{3}\cos(b)}{4}-4\cos(b)^2\cos(c)^4$$

$$-4\cos(c)^2\cos(b)^4+7\cos(b)^2\cos(c)^2-\frac{3}{2}\Bigg)\Bigg)$$

$$-\frac{1}{4}+\frac{\left(-4\sin(b)\cos(b)^3-4\cos(c)^3\sin(c)+5\cos(b)\sin(b)+5\sin(c)\cos(c)\right)\sqrt{3}}{8}\tag{4}$$

$$+\frac{\left(3-4\cos(c)^2\right)\cos(b)^4}{2}+2\sin(c)\cos(c)\sin(b)\cos(b)^3$$

$$+\frac{\left(-16\cos(c)^4+24\cos(c)^2-7\right)\cos(b)^2}{8}+2\left(\cos(c)^2\right.$$

$$\left.-\frac{3}{2}\right)\sin(c)\cos(c)\sin(b)\cos(b)+\frac{3\cos(c)^4}{2}-\frac{7\cos(c)^2}{8}$$

> $h(b,c) := -\frac{1}{4}$

$$+\frac{\left(-4\sin(b)\cos(b)^3-4\sin(c)\cos(c)^3+5\sin(b)\cos(b)+5\cos(c)\sin(c)\right)\sqrt{3}}{8}$$

$$+\frac{\left(-4\cos(c)^2+3\right)\cos(b)^4}{2}+2\sin(c)\cos(c)\sin(b)\cos(b)^3$$

$$+\frac{\left(-16\cos(c)^4+24\cos(c)^2-7\right)\cos(b)^2}{8}+2\sin(b)\left(\cos(c)^2\right.$$

$$\left.-\frac{3}{2}\right)\cos(c)\sin(c)\cos(b)+\frac{3\cos(c)^4}{2}-\frac{7\cos(c)^2}{8}$$

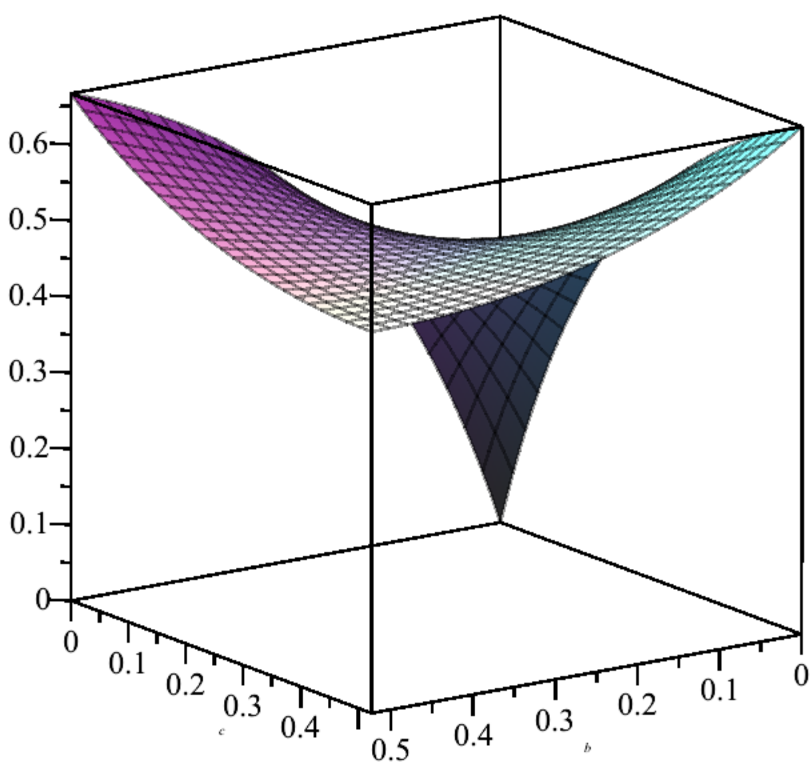
$$h := (b,c)\mapsto -\frac{1}{4}\tag{5}$$

$$+\frac{\left(-4\cdot\sin(b)\cdot\cos(b)^3-4\cdot\sin(c)\cdot\cos(c)^3+5\cdot\sin(b)\cdot\cos(b)+5\cdot\cos(c)\cdot\sin(c)\right)\cdot\sqrt{3}}{8}$$
$$+\frac{\left(-4\cdot\cos(c)^2+3\right)\cdot\cos(b)^4}{2}+2\cdot\cos(c)\cdot\sin(c)\cdot\sin(b)\cdot\cos(b)^3$$
$$+\frac{\left(-16\cdot\cos(c)^4+24\cdot\cos(c)^2-7\right)\cdot\cos(b)^2}{8}+2\cdot\sin(b)\cdot\left(\cos(c)^2-\frac{3}{2}\right)\cdot\cos(c)\cdot\sin(c)$$
$$\cdot\cos(b)+\frac{3\cdot\cos(c)^4}{2}-\frac{7\cdot\cos(c)^2}{8}$$

> $simplify\left(int\left(int\left(\frac{h(b,c)}{\sin\left(b+c+\frac{\pi}{6}\right)^4}, b=0..\frac{\pi}{6}\right), c=0..\frac{\pi}{6}\right)\right)$

$$\frac{1}{4}-\frac{5\ln(3)}{8}+\frac{\pi\sqrt{3}}{24}+\frac{\ln(2)}{2} \tag{6}$$

> $plot3d\left(\frac{h(b,c)}{f(c,b,0)^2}, b=0..\frac{3.14}{6}, c=0..\frac{3.14}{6}\right)$



> $int\left(int\left(\frac{h(b,c)}{f(c,b,0)^2}, b=0..\frac{3.14159265359}{6}, numeric=true\right), c=0..\frac{3.14159265359}{6}, numeric\right.$

$$= true\Big)$$

$$0.1366658305 \tag{7}$$

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